

Small Language Model Research

CSE299 Junior Design — Section 04, Group 03

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I. INTRODUCTION

Small Language Models (SLMs) have emerged as an efficient alternative to large-scale models, enabling deployment under resource-constrained settings while maintaining competitive performance across diverse tasks. This report documents the second weekly update of Group 03, detailing advances across three independent yet complementary research threads. Week 1 established the foundational experimental setup; Week 2 deepens each track with new models, additional benchmarks, and practical application development.

II. WORK COMPLETED

A. SLM Fine-Tuning & Benchmarking

This track challenges conventional instruction-tuning assumptions by replicating established methodology on Phi-4 Mini (2.23B) under resource-constrained conditions. Experiments were conducted on a Kaggle T4 GPU using 4-bit quantization, comparing base-like versus instruct mode across four benchmarks [1].

Key Findings:

- Base-like mode outperformed instruct mode on GSM8K in both zero-shot (13.33% vs. 10.00%) and few-shot (36.67% vs. 16.67%), confirming the paper's core finding on a new model.
- Surprisingly, few-shot examples *degraded* instruct-mode performance on Phi-4 Mini — the opposite effect observed on larger models such as LLaMA-70B and Kimi-K2 in the original study [1].
- Math-500, MedCalc, and Math-Perturb could not be fully evaluated due to a $\LaTeX$ extraction bug and dataset loading errors.

Next Steps: Fix extraction bugs, load all four benchmarks properly, and introduce Gemma-2 2B as a second comparison model.

B. SLM Pipeline Expansion & Evaluation

This week transitioned from tool-level evaluation to deeper pipeline analysis, guided by insights from recent SLM introspection literature [2].

Key Findings:

- Conducted a thorough review of the existing system using the ReflectEvo framework [2], identifying critical gaps in meta-introspection and self-reflection capabilities.

- Evaluated the SLD-VM pipeline on Math and Commonsense QA benchmarks; results unexpectedly weak reasoning capability but exposed gaps in multi-step commonsense tasks.
- Tested the SLD-VM pipeline on the ARC (AI2 Reasoning Challenge) dataset to validate performance across diverse question-answering scenarios.
- Analyzed Phi-4 model performance using varied prompting techniques: zero-shot, few-shot Chain-of-Thought (CoT), and role-based prompting.
- Developed a new pipeline leveraging SGFT (Self-Guided Fine-Tuning) and SCORE to enhance system performance, drawing on self-correcting generation strategies [3].

Next Steps: Complete the New Pipeline and Identify whether Combination of SLM can fill the gap of LLM backing

C. SLM Compression via MoLAE — Extended Experiments

This week expanded compression experiments beyond GPT-2 Medium to validate MoLAE generalizability, and initiated a privacy-sensitive healthcare chatbot application [4].

Key Findings:

- GPT-2 Large was added to the experimental setup alongside GPT-2 Medium, enabling cross-model comparison of MoLAE compression effectiveness [4].
- The Penn Treebank (PTB) dataset was incorporated alongside WikiText-103 to assess perplexity across different linguistic distributions.
- Multi-model and multi-dataset evaluation validates that MoLAE generalizes beyond a single configuration, significantly strengthening the innovation claim.
- Development was initiated on a privacy-first healthcare chatbot powered by compressed SLMs [?], motivated by HIPAA/GDPR compliance requirements.
- **Problem identified:** Current cloud-based LLM healthcare tools transmit patient data externally, posing potential legal violations. **Proposed solution:** Deploy compressed SLMs fully on-device so patient data never leaves the local environment.

Next Steps: Report perplexity results across all model/dataset combinations, finalize chatbot architecture, and begin on-device inference testing.

III. CONCLUSION

Week 2 marked significant progress across all three research tracks. Fine-tuning experiments revealed that instruction-tuned Phi-4 Mini counterintuitively underperforms base-like mode on mathematical benchmarks, challenging standard assumptions about instruction tuning. Pipeline evaluation extended SLD-VM testing to new benchmarks and introduced SGFT+SCORE for improved generation quality. Compression experiments validated MoLAE’s generalizability across models and datasets, while a privacy-first healthcare chatbot was conceptualized and initiated. Week 3 will focus on resolving evaluation bugs, completing benchmark coverage, and advancing cross-member integration.

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